

Hurdle Models and Age Effects in the Major League Baseball Draft

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Abstract

Major League Baseball franchises expend an abundance of resources on scouting in preparation for the June Amateur Draft. In addition to the classic “tools” assessed, another factor considered is age: Younger players may get selected over older players of equal ability because of anticipated development. Additionally, Little League rules in effect until 2006 operated on an August 1–July 31 year meaning that, in their youth, players born on August 1 were the eldest in their cohort. We define a relative age to be how old a player was (in days) when the youngest player in their cohort was born. For example, a player born on July 31 would have a relative age of 1 day, whereas a player born on August 1 would have a relative age of 365 days. We examine the performance of high school (HS) players selected in the June Draft from 1987 to 2011. We find that more relatively old players are selected in the Draft. Conversely, both age and relative age have a significant negative relationship with the odds of reaching the major leagues. Given that a draftee reaches the majors, however, there is no difference based on age or relative age, as measured by games played, wins above replacement, and on-base plus slugging percentage. Had the draft market operated efficiently, neither relative age nor age on draft day would have captured additional variation in performance after controlling for draft position and other factors. We conclude that teams have undervalued both young and relatively young HS players.

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MLB draft, hurdle model, relative age effect, negative binomial regression, underdog effect

Introduction

Since the seminal article by Barnsley, Thompson, and Barnsley (1985), the relative age effect (RAE) has been a hot topic of study in the sports world. Relative age is one's age relative to his or her cohort. For example, in Canadian youth hockey, the age cutoff in peewees is January 1, so the oldest players in a given cohort are those born in January, while the youngest are those born in December. In Little League baseball, the age cutoff was August 1 until 2006, meaning the oldest players in a given cohort were those born in August, while the youngest were those born in July. The RAE is a phenomenon, evident in many sports, where a disproportionate number of participants are observed to be relatively older. Previous studies have observed the RAE to exist in both hockey (Barnsley, Thompson, & Barnsley, 1985) and baseball (Thompson, Barnsley, & Stebelsky, 1991). In addition to the initial studies on hockey and baseball, this same effect has been deeply chronicled in minor league and professional hockey, and soccer, as well as observed in American football (Daniel & Janssen, 1987; Glamser & Marciani, 1992; Stanaway & Hines, 1995), handball (Ryan, 1989), swimming (Baxter-Jones, 1995; Ryan, 1989), tennis (Baxter-Jones, 1995; Dudink, 1994), and volleyball (Ryan, 1989).

We make a distinction in this article between age and relative age. Age is a draft prospect's age on Major League Baseball (MLB) draft day, while relative age deals with Little League cutoff dates. In hockey, since the playing year is the same as the calendar year, there is no distinction needed since the oldest/youngest players are also the relatively oldest/youngest. In baseball, however, a player can be relatively old (e.g., born in August), but young for his draft class (e.g., a high school [HS] player drafted under the age of 18) or vice versa. Jazayerli (2011) claims that young HS hitters for their draft class tend to outperform their older counterparts in terms of future production. Other potential market inefficiencies in the MLB Draft have been investigated, including whether certain positions have been overvalued (Burger & Walters, 2009; Salaga, 2012), whether college players are better choices than HS players (Bradbury, 2011; Burger & Walters, 2009; Lewis, 2003; Salaga, 2012; Spurr, 2000), whether certain teams are better at evaluating talent than others (Spurr, 2000), and whether the geographic area a player is drafted out of has a significant effect on career performance (Salaga, 2012).

Some recent research has focused on an RAE *reversal*, where relatively young individuals tend to outperform their relatively old counterparts at higher levels of competition (Deaner, Lowen, & Copley, 2013; Gibbs, Jarvis, & Dufur, 2011). One potential reason given for this finding is the "underdog effect," where relatively

young players have had to consistently work harder to compete with their relatively old counterparts who were more physically and mentally mature. Another potential explanation is that, given that a relatively young player does not quit at a young age, he or she must have been good enough to compete with the relatively old players at that age, suggesting that the relatively young player was more naturally talented.

This article adds to the literature by examining the effects of both age and relative age on MLB performance. Prior literature has only considered relative age with respect to the birth month distribution of players who had already reached the majors. To the best of our knowledge, no prior baseball research has examined the birth month distribution of draftees, the success rate at which these draftees reach the majors, or their performance once in MLB. Gibbs, Jarvis, and Dufur (2011) and Deaner, Lowen, and Cogley (2013) explore RAEs on performance in hockey, where the cutoff date aligns with the calendar year. In baseball, age and relative age are distinct, and we consider their effects simultaneously for the first time. In particular, we explore the relationships of age and relative age for players drafted out of HS with baseball career performance, using four different performance metrics (whether or not the player reached the major leagues, games played in MLB, career wins above a replacement player [WAR], and career on-base plus slugging percentage [OPS] for nonpitchers only). The second section describes the data and methods used in our analysis, the third section presents the results of our study, and in the fourth section, our findings are discussed. The fifth section provides some concluding remarks.

Data and Methods

Baseball-Reference.com (MLB June Amateur Draft, 2013) and TheBaseballCube.com (Amateur Draft History, 2013) have in-depth information on every player selected in the June Draft. We gathered data for all players drafted in the Top 50 rounds of the June Draft from 1987 to 2011. We gathered information over this period, as 1987 was the first year in which the June Draft was the only draft where players were selected from HSs and colleges in the United States and territories. Before 1987, other drafts included the June Secondary, January, and January Secondary Drafts.

We collected data for players through the 2011 draft, as very few players drafted in 2012 and 2013 had reached the majors at the time of our analysis. All performance data are as of June 12, 2013, and gathered from Baseball-Reference.com. The relevant variables are Position at the time of draft, WAR, OPS, games, and type of school from which drafted (college, HS, junior college, or other). The relevant variables collected from TheBaseballCube.com are draft year, overall pick number, draft round, birthday, birth month, and birth year.

We created a “last time drafted variable” (lastDraft) that is a dummy variable, where 1 indicates that, as of the 2011 Draft, this was the last time a player was drafted in the Top 50 rounds. We created lastDraft in order to control for players who

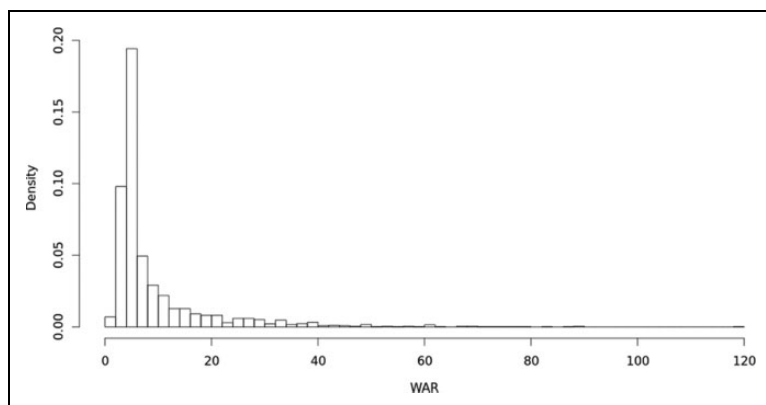


Figure 1. WAR for players who played at least one game in the majors and were drafted in the Top 5 rounds. WAR = wins above a replacement player.

did not sign with their drafting team and decided to attend college. Previous research has shown that this can be a valuable decision in terms of expected lifetime earnings (Winfree & Molitor, 2007). We calculated age (in years) to express how old the draft pick was on draft day. Additionally, we defined relative age (in days) to be how old a player was when the youngest possible player in their Little League cohort was born, given an August 1–July 31 year. A relative age of 1 day represents the youngest player in a cohort (i.e., born on July 31) and a relative age of 365 days represents the oldest player in a cohort (i.e., born on August 1). Age and relative age showed very little association in our data, with a correlation coefficient of approximately .10 for all draftees in the Top 50 rounds.

Career Performance Measures

A player's WAR is intended to measure the additional number of wins that the player's team accumulates over a replacement-level player. A replacement-level player is defined as one whom the team could sign for the minimum salary. Nevertheless, a few major league players end their careers with negative WAR values. A negative WAR is philosophically problematic in the sense that a player who made the majors could be considered as having a worse career than another who did not make the majors. Therefore, we shifted WAR, so that all values are positive. This new measure can be interpreted as the wins above the worst observed player in our sample. For simplicity, we continue to refer to this measure as WAR.

Previous research has indicated that young players have been undervalued in the draft in terms of future WAR production using ordinary least squares (OLS) regression (Jazayerli, 2011). We found that heteroscedasticity was present and, given an OLS framework, we note that the error term is not normally distributed (model diagnostics not shown). Alternatively, we observe that WAR is more appropriately

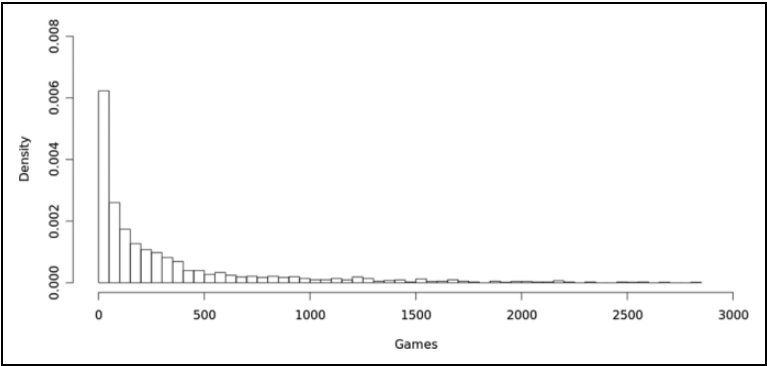


Figure 2. Games played for players who played at least one game in the majors and were drafted in the Top 5 rounds.

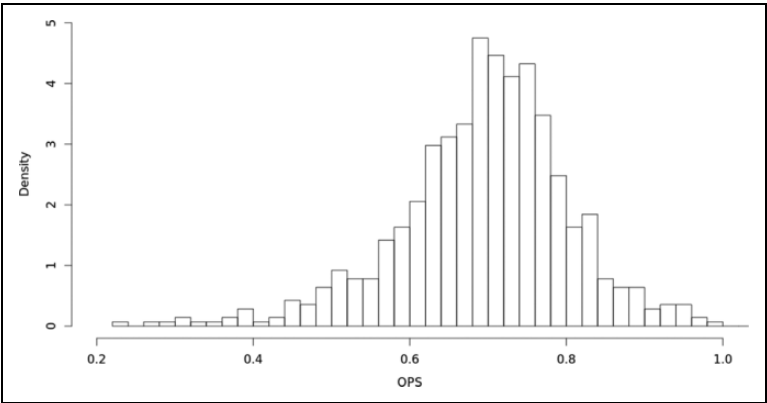


Figure 3. OPS for Top 5 round draftees selected as position players who played more than 20 games in the majors. OPS = on-base plus slugging percentage.

modeled using a lognormal distribution (see Figure 1). We also believe that the process is better modeled as a two-stage process, where we first estimate the relationships of age and relative age with reaching the major leagues, given that a player is drafted. We refer to this first stage as the hurdle, since a player must get over a barrier to reach the major leagues and accumulate statistics. Then, given that a player has reached the majors, we estimate the relationships of age and relative age with production at the major league level using a generalized linear model. In addition to WAR, we use two other performance metrics: games played in MLB (which we model as a zero-inflated negative binomial; see Figure 2) and OPS for hitters (which we model as a normal; see Figure 3).

We will focus our methodology on games played in MLB, as this is the most interesting of the three cases. WAR and OPS are modeled with continuous distributions. Thus, once the hurdle is overcome, estimation proceeds in a standard fashion. Games played is a count variable with excess zeros. We note that these excess zeros are structural, that is, the only way a player could have zero games played is if he did not reach the major leagues. In actuality, it is possible for a player to be on a major league roster without ever appearing in a game, but this is very unlikely. Since there is only one source of zeros it is appropriate to use a hurdle model, as opposed to a zero-inflated model, which does not distinguish between structural and nonstructural zeros. Moreover, there is overdispersion in the data, so it is appropriate to use negative binomial regression, as opposed to Poisson regression.

In summary, we use logistic regression to model the probability of reaching the majors, followed by a truncated negative binomial regression to model the number of games played at the MLB level, given that a player has reached the majors. Formally, the density of the hurdle model can be written as follows:

$$f_{\text{hurdle}}(y; x, z, \beta, \gamma) = \begin{cases} f_{\text{zero}}(0; z, \gamma) & \text{if } y = 0 \\ (1 - f_{\text{zero}}(0; z, \gamma)) \frac{f_{\text{count}}(y; x, \beta)}{(1 - f_{\text{count}}(0; x, \beta))} & \text{if } y > 0, \end{cases} \quad (1)$$

where y represents games played, f_{zero} is the binomial probability function, and f_{count} is the negative binomial probability function (Zeileis, Kleiber, & Jackman, 2012). The covariates in the count model, x , can be different from the covariates in the binomial model, z , if the factors influencing getting over the hurdle are different from those influencing the count once over the hurdle. In our analyses, however, we use the same covariates in each model, so $x = z$. The model parameters, β and γ , are estimated by maximum likelihood, where the hurdle and count components can be maximized separately. Since f_{count} is the negative binomial probability function, an overdispersion parameter is estimated which distinguishes this from Poisson regression. This parameter is also estimated by maximum likelihood.

Due to the truncated zeros in the count model, interpretation of the coefficients is difficult compared with standard count regression (Sampford, 1955). We now elaborate on this point briefly, since much of the applied work that employs these models does not make this clear. The negative binomial distribution with parameters k and r can be written as follows:

$$P(r) = \frac{(k + r - 1)!}{(k - 1)!r!} \frac{p^r}{(1 + p)^{k+r}} \quad (r = 0, 1, \dots; p > 0; k \geq 1). \quad (2)$$

We observe that the expected value using the parametrization in Equation 2 is kp . We also note that

$$P(0) = \frac{1}{(1 + p)^k}. \quad (3)$$

Thus, we must divide Equation 2 by Equation 3 to find the truncated negative binomial distribution used in Equation 1. First, we let

$$\omega = \frac{1}{1+p} \quad \text{and} \quad \eta = 1 - \omega.$$

We find that the truncated negative binomial distribution, $P_t(r)$, has the form

$$P_t(r) = \frac{\omega^k}{1 - \omega^k} \frac{(k+r-1)!}{(k-1)!r!} \eta^r \quad (r = 1, 2, \dots; k \geq 1; \omega, \eta \in (0, 1)), \quad (4)$$

with expected value $\frac{k\eta}{\omega(1-\omega^k)}$.

Negative binomial count models are built for the parameter k . In standard negative binomial regression, we observe that if $k = \exp(\mathbf{X}\beta)$, then $E(R) = \exp(\mathbf{X}\beta)p$, which represents a constant multiplicative change in the mean by $\exp(\beta_i)$ for 1 unit increases of $x_i \in \mathbf{X}$. However, the truncated negative binomial model does not possess this property, as

$$E(R_t) = \frac{\exp(\mathbf{X}\beta)\eta}{\omega(1 - \omega^{\exp(\mathbf{X}\beta)})}, \quad (5)$$

which we can easily verify has a rate of change with respect to $x_i \in \mathbf{X}$ that is not independent of the values of x_i . Consequently, there is no simple interpretation of β_i . In our subsequent tables of results, we report coefficient values, and for simplicity, we discuss $\exp(\beta_i)$ as an approximation of the multiplicative change in the mean for 1 unit increases of x_i .

Results

Distribution of Birth Month for Drafted Players

Figure 4 presents a histogram for the birth month distribution of all Top 50 round MLB draftees from 1987 to 2011. Moreover, Table 1 illustrates the observed frequencies of birth months for draftees compared with theoretical frequencies based on uniform births and observed births in the United States from 1995 to 2002 (James, 2005). The chi-square goodness-of-fit test reveals that the observed distribution of the birth month of draftees was significantly different from both the uniform distribution and the theoretical distribution, given observed births in the United States ($p < 2.2\text{e}-16$).

Probability of Reaching the Major Leagues

The results of the logistic regression portion of the hurdle negative binomial models for the probability of reaching the majors (for players drafted in the Top 5 and Top 10 rounds) are reported in Table 2. Although we do not report the coefficients for draft year, we use indicator variables for each year considered to control for changes

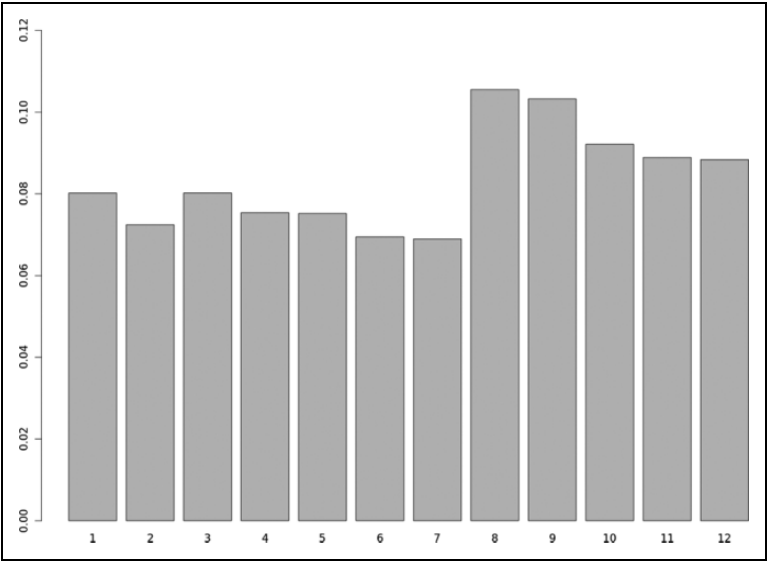


Figure 4. Birth month distribution for Top 50 round draftees from 1987 to 2011, with proportion shown on vertical axis.

Table 1. Birth Month Frequencies for Top 50 Round Draftees From 1987 to 2011.

Month	Absolute Frequency	Relative Frequency	Theoretical frequency (Uniform)	Theoretical frequency (observed United States births)
January	2,582	0.0802	0.085	0.0815
February	2,332	0.0724	0.077	0.0761
March	2,583	0.0802	0.085	0.0835
April	2,427	0.0754	0.082	0.0801
May	2,421	0.0752	0.085	0.0844
June	2,238	0.0695	0.082	0.0830
July	2,220	0.0689	0.085	0.0880
August	3,398	0.1055	0.085	0.0888
September	3,324	0.1032	0.082	0.0865
October	2,967	0.0921	0.085	0.0851
November	2,863	0.0889	0.082	0.0799
December	2,845	0.0884	0.085	0.0831

across draft years. We see that both relative age and age have negative and significant relationships with the odds of making the majors. Over the Top 5 rounds, the odds of a player making the majors are 69% of those for a player 1 year younger on draft day ($p = .0038$). We also find that the odds of a player making the majors are 89% of those for a player 100 days relatively younger ($p = .0326$). We use

Table 2. Logistic Regression Portion of Hurdle Model Using, as the Response Variable, an Indicator of Whether a Draftee Reached MLB: (i) Top 5 Rounds, (ii) Top 10 Rounds, (iii) Top 5 Rounds With lastDraft, (iv) Top 10 Rounds With lastDraft.

Explanatory variable	(i)	(ii)	(iii)	(iv)
Intercept	5.903* (2.339)	5.366** (1.814)	5.953* (2.342)	5.515** (1.818)
Relative age	-0.0011* (0.0005)	-0.0009* (0.0004)	-0.0011* (0.0005)	-0.0009* (0.0004)
Age (in years)	-0.3693** (0.1276)	-0.3615*** (0.0988)	-0.3667** (0.1278)	-0.3568*** (0.0990)
$1/\sqrt{\text{pick}}$	5.317*** (0.6432)	7.261*** (0.6091)	5.362*** (0.6484)	7.454*** (0.6180)
Pitcher	0.2423* (0.1073)	0.3183*** (0.0886)	0.2408* (0.1073)	0.3165*** (0.0887)
lastDraft	—	—	-0.1127 (0.1827)	-0.3141* (0.1303)
Observations	1,807	2,999	1,807	2,999
Generalized R^2	.271	.231	.271	.233

Note. MLB = Major League Baseball.

Natural logarithm of odds ratios reported, with standard errors in parentheses. Generalized R^2 values due to Cox and Snell (1989).

*Implies statistical significance at the 5% level. **Implies statistical significance at the 1% level. ***Implies statistical significance at the 0.1% level.

$1/\sqrt{\text{pick}}$ as our transformation of overall draft position, as we note that the relationship between performance metrics and draft position is not linear (see Figures 5 and 6). Additionally, we note that a 0.1 unit increase in $1/\sqrt{\text{pick}}$ (e.g., the 100th pick to the 25th pick) was associated with a 70% increase in the odds of reaching MLB.

We created a pitcher indicator variable to control for differences across performance metrics between hitters and pitchers. The coefficient on pitcher is positive, suggesting that they have been more likely to reach the majors than hitters. Some teams may call up many pitchers in a year due to the greater risk of injury for pitchers.

The results do not change substantially for the Top 10 rounds (see Table 2 (ii)). Furthermore, the results for the Top 50 rounds (not reported) are similar to the Top 10 rounds, suggesting that our findings are robust to changes in the draft rounds considered. Our main results are robust to the inclusion of lastDraft (see Table 2 (iii) and (iv)).

Career Games Played in the Major Leagues

Using total games played as a measure of career success, we find consistent results across draft rounds and note that the inclusion of lastDraft in the models does not alter our main findings (see Table 3). We shall examine the models excluding lastDraft (see Table 3 (i) and (ii)). For Top 5 round draftees, both age and relative age

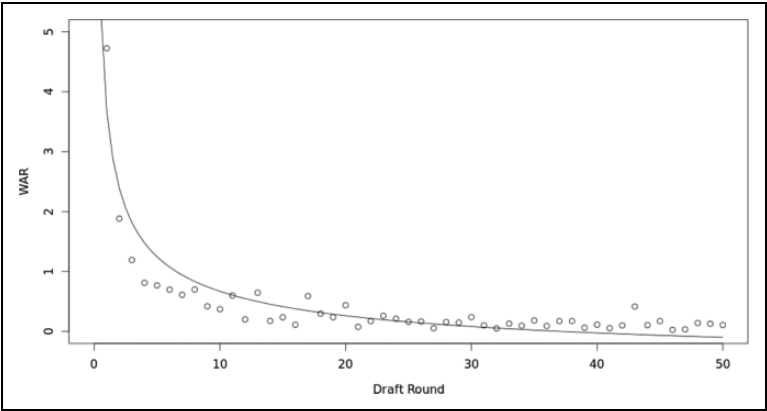


Figure 5. WAR by draft round ($1/\sqrt{\text{pick}}$ curve shown). WAR = wins above a replacement player.

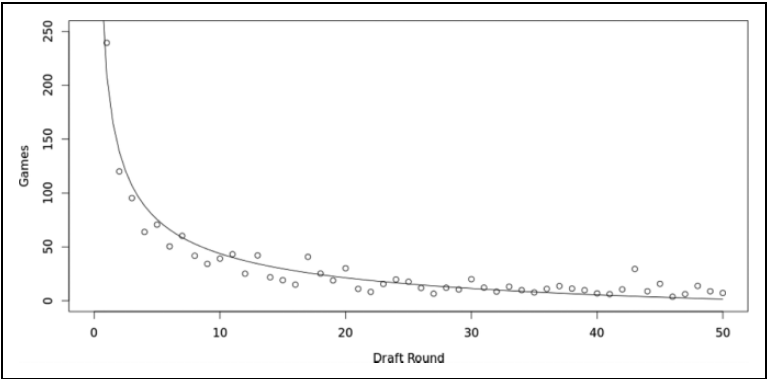


Figure 6. Games played by draft round ($1/\sqrt{\text{pick}}$ curve shown).

are negatively, but statistically insignificantly, correlated with total games played. We note that a player 1 year older on draft day played in 92% ($p = .5076$) of the games compared to the younger player over the course of their career. Additionally, a player 100 days relatively older played in 94% ($p = .2239$) of the career games compared to the younger player. For a 1,000 game career, this difference is approximately 56 games, roughly one third of a full 162 game season.

Furthermore, over the Top 5 rounds, $1/\sqrt{\text{pick}}$ is positively correlated ($p < .001$) with games played (see Table 3 (i)). The results indicate that a 0.1 unit increase in $1/\sqrt{\text{pick}}$ is associated with a 19% increase in games played. This means that, holding all else constant, a player drafted with the 25th overall pick played in 19% more games than a player selected 100th overall. If the 100th pick played in 1,000 career

Table 3. Negative Binomial Count Portion of Hurdle Model Using, as the Response Variable, Career Games Played in MLB: (i) Top 5 Rounds, (ii) Top 10 Rounds, (iii) Top 5 Rounds With lastDraft, (iv) Top 10 Rounds With lastDraft.

Explanatory variable	(i)	(ii)	(iii)	(iv)
Intercept	7.408*** (2.183)	8.163*** (1.911)	7.317*** (2.192)	8.051*** (1.918)
Relative age	-0.0006 (0.0005)	-0.0007 (0.0004)	-0.0006 (0.0005)	-0.0007 (0.0004)
Age (in years)	-0.0784 (0.1183)	-0.1167 (0.1048)	-0.0774 (0.1184)	-0.1162 (0.1050)
$1/\sqrt{\text{pick}}$	1.749*** (0.3739)	1.744*** (0.3544)	1.724*** (0.3767)	1.677*** (0.3576)
Pitcher	-1.208*** (0.1005)	-1.238*** (0.0898)	-1.209*** (0.1005)	-1.245*** (0.0900)
lastDraft	—	—	0.0800 (0.1728)	0.1350 (0.1301)
Log(θ)	-0.3248*** (0.0592)	-0.3876*** (0.0529)	-0.3245*** (0.0592)	-0.3863*** (0.0529)
Observations	1,807	2,999	1,807	2,999
Generalized R^2	.271	.231	.271	.233

Note. MLB = Major League Baseball.
Natural logarithm of approximate multiplicative change in games played reported, with standard errors in parentheses. Generalized R^2 values due to Cox and Snell (1989).
*Implies statistical significance at the 5% level. **Implies statistical significance at the 1% level. ***Implies statistical significance at the 0.1% level.

games, then the 25th pick played in 1,191 games, more than a full season of additional games. Not surprisingly, being a pitcher has a negative relationship ($p < .001$) with a player's total games played.

Recall that, for the hurdle models, we chose to use negative binomial regression as opposed to Poisson regression due to anticipated overdispersion (greater than expected variability). This was an appropriate choice, as the overdispersion parameter, Log(θ), is significant ($p < .001$; see Table 3). The results for the Top 10 rounds of the draft (see Table 3) and Top 50 rounds (not reported) differ only inconsequentially.

Career WAR in the Major Leagues

We also use career WAR to estimate the relationships between age and relative age and MLB performance, given that a drafted player made the majors. For the Top 5 round draftees (see Table 4 (i)), a player 100 days relatively older than another produced 95% of the younger player's WAR ($p = .106$). Furthermore, a player 1 year older than another on draft day produced 89.28% of the WAR of the younger player ($p = .141$). We note that a 0.1 unit increase in $1/\sqrt{\text{pick}}$ was correlated with a 19%

Table 4. Lognormal Regressions Using, as the Response Variable, Career WAR: (i) Top 5 Rounds. (ii) Top 10 Rounds, (iii) Top 5 Rounds With lastDraft, (iv) Top 10 Rounds With lastDraft.

Explanatory variable	(i)	(ii)	(iii)	(iv)
Intercept	3.583* (1.419)	2.806* (1.092)	3.625* (1.422)	2.796* (1.092)
Relative age	-0.0005 (0.0003)	-0.0005* (0.0003)	-0.0005 (0.0003)	-0.0005* (0.0003)
Age (in years)	-0.1134 (0.0771)	-0.0666 (0.0596)	-0.1133 (0.0771)	-0.0672 (0.0597)
$1/\sqrt{\text{pick}}$	1.723*** (0.2170)	1.618*** (0.1970)	1.736*** (0.2189)	1.606*** (0.1999)
Pitcher	-0.0737 (0.0631)	-0.0889 (0.0541)	-0.0729 (0.0631)	-0.0894 (0.0541)
lastDraft	—	—	-0.0482 (0.1107)	0.0286 (0.0801)
Log(scale)	-0.2696*** (0.0283)	-0.2876*** (0.0247)	-0.2697*** (0.0283)	-0.2877*** (0.0247)
Observations	624	818	624	818
Generalized R^2	.147	.120	.147	.120

Note. WAR = wins above a replacement player.

Natural logarithm of multiplicative change in WAR reported, with standard errors in parentheses. Generalized R^2 values due to Cox and Snell (1989).

*Implies statistical significance at the 5% level. **Implies statistical significance at the 1% level. ***Implies statistical significance at the 0.1% level.

increase in WAR ($p < .001$). Furthermore, pitchers produced 93% of the WAR that position players produced ($p = .242$). Interestingly, for Top 10 round draftees (see Table 4 (ii)), we see that a player 100 days relatively older than another produced 95% of the WAR of the younger player, which attains significance at the 5% level ($p = .0436$). The remainder of the results were not substantively different.

Career OPS in the Major Leagues

Thus far, we have only examined career *cumulative* statistics. Here, we take a closer look at a career *average* statistic, OPS. We perform weighted regressions, using the number of career at bats as weights. The OPS data are well modeled by a normal distribution (see Figure 3). The results of these regressions are reported in Table 5. They are not substantively different across draft rounds. For Top 5 round draftees, a player produced .00038 less OPS ($p = .9303$) compared to a player 100 days relatively younger. This is a minuscule amount baseball wise, as OPS is only measured up to three decimal places. A player 1 year older than another on draft day produced .0084 less OPS ($p = .4646$) on average. A 0.1 unit increase in $1/\sqrt{\text{pick}}$ was associated with a 0.012 unit increase in OPS ($p < .001$).

Table 5. Weighted (by At Bats) Ordinary Least Squares Regressions Using, as the Response Variable, OPS: (i) Top 5 Rounds, (ii) Top 10 Rounds, (iii) Top 5 Rounds With lastDraft, (iv) Top 10 Rounds With lastDraft.

Explanatory variable	(i)	(ii)	(iii)	(iv)
Intercept	0.8897*** (0.2101)	0.9103*** (0.1872)	0.9268*** (0.2067)	0.9295*** (0.1862)
Relative age	-0.000004 (0.00004)	-0.00005 (0.00004)	-0.00002 (0.00004)	-0.00005 (0.00004)
Age (in years)	-0.0084 (0.0115)	-0.0091 (0.0103)	-0.0075 (0.0113)	-0.0087 (0.0103)
$1/\sqrt{\text{pick}}$	0.1215*** (0.0185)	0.1116*** (0.0171)	0.1298*** (0.0184)	0.1194*** (0.0173)
lastDraft	—	—	-0.0545*** (0.0159)	-0.0307* (0.0128)
Observations	322	405	322	405
R ²	.295	.218	.323	.230

Note. OPS = on-base plus slugging percentage.
 Additive change in OPS reported, with standard errors in parentheses.
 *Implies statistical significance at the 5% level. **Implies statistical significance at the 1% level. ***Implies statistical significance at the 0.1% level.

Discussion

We find that there exists strong evidence for the RAE in the MLB draft. That is, we see that the majority of draftees have birthdays in the first few months of the Little League year. This result is consistent across draft rounds.

We note that for HS players there appears to be a RAE *reversal* on the odds of reaching the major leagues. It is possible that this is due to what has been described as the “underdog effect,” where the relatively young players have had to work harder their whole lives to be as good as the relatively older players (Deaner et al., 2013; Gibbs et al., 2011). This propels the relatively younger players to reach higher levels.

Another possible explanation is more related to the RAE itself. Take, for example, a relatively young player who struggles to compete against his relatively older counterparts; he may drop out of playing ball at a young age. This is one of the proposed explanations for the RAE. But take that same relatively young player and suppose he enjoys success at a young age playing against his relatively older peers. This player will be less likely to drop out. Given that he enjoyed success as a relatively younger player, it is likely that he is more naturally talented. Thus, he continues to play and is more talented, which in turn indicates that he is more likely to make the majors.

The question arises as to why this does not continue at the major league level. As we observe from our games, WAR, and OPS regressions, there does not appear to be an RAE reversal at the major league level. This means that, given that a player has

reached the majors, he is of approximately equal ability as all other players. Why would this be the case? We will explore this issue shortly.

Furthermore, not only do relatively young players make the majors more often, but absolutely young players do too. This result can be more easily explained. If a player is drafted as an 18-year-old out of HS and begins his pro-career in rookie ball—a fairly regular occurrence—he is likely to be young for his level. If he enjoys success at this level, he can be promoted to A ball. Now, take, for example, the same 18-year-old who struggles in rookie ball. Since he is young for his level, he may be given an opportunity the following year to try to prove himself worthy by repeating the level (i.e., getting a second chance). If another player were 19 years old when drafted, placed in rookie ball, and struggled, the team would be less willing to let that player have another opportunity. He may quickly move down the organizational depth chart at his position. Thus, a player who is younger when drafted may be given extra opportunities in the minors, enabling them to reach the majors more often.

As with relative age, we see that absolute age has no significant relationship with performance once a player has reached the majors. Again we ask, why is this the case? Our results seem to indicate that there is an ability threshold that a player must achieve in order to reach the majors. Once players have reached MLB, differences in age and relative age do not signify differences in performance.

Conclusion

We find that the RAE does indeed exist in the MLB Draft. Further, we can conclude that even though more relatively older players reach the major leagues, this is only due to the larger proportion of relatively old players drafted. Among drafted players, we see a statistically significant *negative* impact on reaching the majors for the relatively old, suggesting evidence of an RAE reversal. Once players reach the majors, we find no evidence that relatively old players perform differently than relatively young players, and this result holds for absolute age as well. This is analogous to other research which suggests that RAEs disappear at the highest levels of athletics. For example, Addona and Yates (2010) find that even though the RAE was present in the National Hockey League across all positions, the RAE was not evident for hall of fame players.

There are many possible avenues for future work on the RAE in baseball. For instance, one could examine whether the RAE exists for international players in MLB or in other professional leagues around the world. Another possible research question deals with the age of international signees. In particular, whether older players signed out of Latin American countries tend to outperform their signing bonus. One difficulty with the latter research question is that international players may lie about their age in order to appear more desirable to major league teams. A third potential line of research addresses the change in the Little League year.

As of 2006, Little League has been operating on a May 1–April 30 year. This provides a natural experiment that can be exploited to see whether changes in the birth month distribution of draftees occur. That is, we might expect an influx of May, June, and July draftees over the course of the next few years.

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